

An Appraisal Theoretic Approach to Modelling Affect Flow in Conversation Corpora

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Introduction

- Emotion modelling in conversation is a key task for understanding and “knowing when to” respond to emotional cues.
- But emotion annotations in existing corpora are inconsistent across domains (e.g., Ekman, Plutchik, VAD).
- These inconsistencies hinder generalization, evaluation, and comparison between models.
- We propose using **Appraisal Theory** as a generalizable, interpretable affect representation.

Some Existing Approaches to Emotion Modelling

Dataset Name	Type	Annotation	Layer	Domain	Size
EmpatheticDialogues	Conversations	32 emo cat	dialogue	General	24.8K
Epitome	Reddit	3 2pt dim	response	Mental Health	10.1K
DailyDialog	Conversations	5 emo cat	turn	General	13.1K
EmoWOZ	Conversations	3 senti + 7 emo cat	turn	Task Oriented	11.4K
WASSA	Conversations	3 5pt dim	response	News	12.6K
Condolence	Reddit	Distress/support	comment	Online Support	14.1M
ALoE	Reddit	Empathy levels	post	Mental Health	10K
ESConv	Conversations	Support strategies	turn	Mental Health	1K
MELD	Conversations (M.P)	3 senti + 7 emo cat	turn	Movies	13.7K
IEMOCAP	Conversations	3 5pt dim + 7 emo cat	turn	Multimodal	10K
Twitter-Customer	Tweet-Response	3 5pt dim + 7 emo cat	turn	Customer Service	9K

Appraisal Theory

- Appraisal Theory: Emotions are the result of evaluating events along cognitive dimensions.
- Dimensions determined from outcomes/objectives: relevance, implication, coping, and normative significance.
- Enables us to model emotions as **structured cognitive states**, not black-box labels.
- These are more stable across domains and interpretable across contexts.

Research Hypotheses

- **H1:** Appraisal-based emotion estimation aligns with existing emotion labels across corpora.
- **H2:** Changes in appraisal dimensions across conversation turns reflect **affect flow**, i.e. the evolution of emotion in interaction.
- To test these, we introduce **AppraisePLM**, a multitask model trained on the **crowd-EnVENT** dataset

- 6,600 event descriptions annotated with:
 - 21 cognitive appraisal dimensions based on the four objectives (e.g. pleasantness, certainty, own-control, etc.)
 - 13 emotion categories based on Smith and Ellsworth (1985) but with more appraisal dimensions
- “Their experimental set-up allowed them to measure if emotions and appraisals can be recovered purely from text and provides a human baseline to judge a model’s performance measures” (Troiano et. al. 2022).
- Used for training both appraisal regression and emotion classification.

- Multitask model:
 - Uses a pretrained language model (PLM) to represent the conversation turn
 - Regression: Estimate 21 appraisal dimensions from text
 - Classification: Use text + appraisals to classify emotion (13 emotion classes including no-emotion)
- Built on transformer encoders (RoBERTa, DeBERTa, T5); DeBERTa performed best.
- Architecture includes multihead attention for each task.

Performance on crowd-EnVENT

- DeBERTa + appraisal-enhanced classification achieves $F1 = 0.71$ (vs 0.60 baseline).
- Appraisal regression yields $MSE = 1.08$, on par with state-of-the-art.
- Adding appraisal info improves classification significantly (~ 0.11 F1 gain).

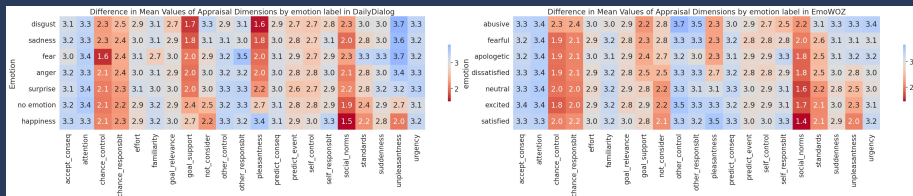
Model	$T \rightarrow A$	$T \rightarrow E$	$T + A \rightarrow E$
	MSE $\downarrow$	F1 $\uparrow$	F1 $\uparrow$
Troiano et. al. (2022)	1.97	0.59	0.60
RoBERTa-large	1.62	0.59	0.67
T5-large	1.12	0.61	0.66
MPNet-base	1.08	0.64	0.70
DeBERTa-large	1.08	0.66	0.71

Cross-Corpus Evaluation

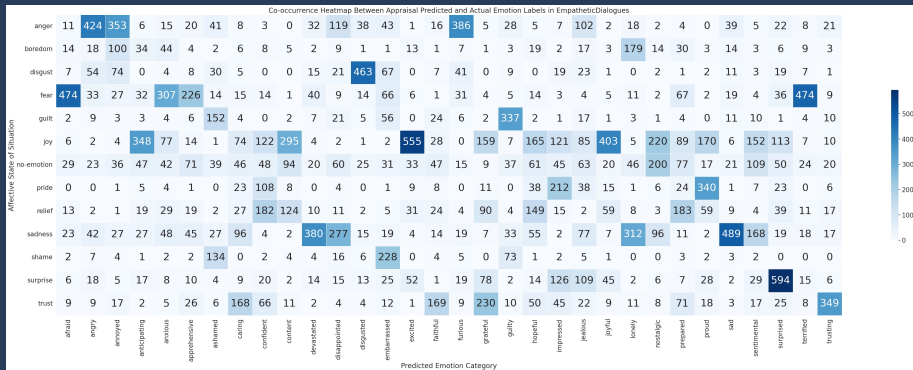
- AppraisePLM was zero-shot evaluated to examine the utility of appraisals estimates from crowd-EnVENT
 - DailyDialog (Li et. al., 2017): General-domain chit-chat
 - EmpatheticDialogues (RAshkin et. al., 2019): Situation-grounded Empathetic Conversations
 - Epitome (Sharma et. al., 2020)Tw: Reddit mental health support
 - EmoWoZ (Feng et. al., 2022): Emotion-annotated task-oriented dialogue
- Estimate appraisal values and predict emotion from *each turn*.

Appraisal Estimates Across Corpora

- Appraisal predictions mirror expected emotional valence (e.g., joy = high pleasantness).
- Corpus-specific norms influence appraisal distribution (e.g., EmoWOZ = flatter profiles).
- Suggests model captures emotion meaningfully beyond training domain.



Case Study: EmpatheticDialogues



Case Study: EPITOME

- EPITOME uses empathy ratings: Reaction, Interpretation, Exploration (0–2) (Sharma et. al., 2020).
 - **Emotional Reaction:** Expressing emotions such as warmth, compassion, and concern, experienced by peer supporter
 - **Interpretation:** Communicating an understanding of feelings and experiences inferred from the seeker's post
 - **Exploration:** Improving understanding of the seeker by exploring the feelings and experiences not stated
- AppraisePLM captures these via appraisal deltas.

Case Study: EPITOME (cont.)

		Difference in Mean Values of Appraisal Dimensions by ER_level																				
ER_level	0	1.5	2.9	-0.7	-0.5	1.6	1.0	1.0	0.8	-1.9	1.5	4.7	2.7	0.0	-0.0	0.3	-2.1	-0.7	-0.5	-1.2	-2.4	1.0
	1	3.0	2.6	-1.7	-2.0	0.9	1.3	2.4	2.7	-6.2	5.2	10.5	10.4	0.1	-2.0	-0.8	-4.5	-0.7	-2.2	-3.9	-13.7	1.0
	2	0.2	0.9	-0.3	-0.8	-0.3	1.3	3.1	0.8	-2.5	4.1	7.4	7.2	0.7	-0.8	0.4	-4.5	-1.1	-1.4	-4.2	-6.1	1.2
		accept_conseq	attention	chance_control	chance_responsibit	effort	familiarity	goal_relevance	goal_support	not_consider	other_control	other_responsibit	pleasantness	predict_conseq	predict_event	self_control	self_responsibit	social_norms	standards	suddenness	unpleasantness	urgency

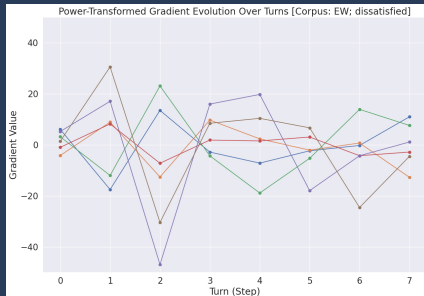
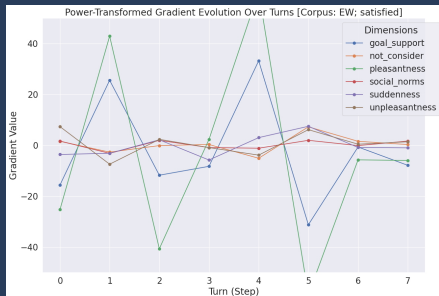
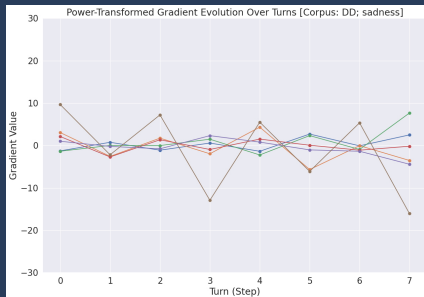
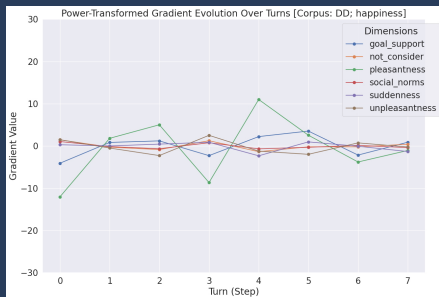
		Difference in Mean Values of Appraisal Dimensions by IP_level																				
IP_level	0	1.8	2.6	-1.2	-1.2	-0.8	1.9	0.9	1.9	-4.5	5.4	11.3	7.7	0.9	-1.1	-0.3	-5.9	-0.5	-0.4	-2.3	-9.5	-0.5
	1	2.0	4.2	-1.0	-0.4	2.5	2.2	2.4	1.5	-3.0	2.5	7.2	6.6	0.9	0.9	2.4	-0.5	-1.2	-1.6	-3.2	-3.4	1.5
	2	1.9	2.7	-0.7	-0.7	3.7	0.1	2.1	0.6	-1.6	-0.6	0.5	1.9	-0.9	-0.2	0.2	0.5	-1.0	-1.7	-1.9	-1.6	2.9
		accept_conseq	attention	chance_control	chance_responsibit	effort	familiarity	goal_relevance	goal_support	not_consider	other_control	other_responsibit	pleasantness	predict_conseq	predict_event	self_control	self_responsibit	social_norms	standards	suddenness	unpleasantness	urgency

		Difference in Mean Values of Appraisal Dimensions by EX_level																				
EX_level	0	2.1	1.9	-0.8	-0.8	1.2	1.3	1.2	1.6	-3.5	1.8	4.5	5.3	-0.1	-0.1	0.5	-1.8	-1.1	-1.6	-3.2	-5.5	1.1
	1	2.6	7.6	-2.9	-3.7	1.8	3.0	3.3	1.7	-4.1	10.5	19.4	7.9	2.2	-3.0	-1.8	-8.8	0.9	1.0	4.8	-12.3	-0.3
	2	-0.0	6.5	-1.6	-1.4	2.0	-0.4	3.2	-0.5	-0.4	6.7	16.5	3.4	0.8	-3.6	-2.8	-9.4	1.2	2.4	3.3	-6.6	0.8
		accept_conseq	attention	chance_control	chance_responsibit	effort	familiarity	goal_relevance	goal_support	not_consider	other_control	other_responsibit	pleasantness	predict_conseq	predict_event	self_control	self_responsibit	social_norms	standards	suddenness	unpleasantness	urgency

Modelling Affect Flow

- Track how appraisal dimensions evolve across turns in a dialogue.
- Compute gradients per dimension per turn: e.g., pleasantness increase/decrease.
- Compare flow patterns across emotions and corpora.
- Emotion-specific shifts observed: anger $\rightarrow$ neutral (pacification); joy $\rightarrow$ stable.

Case Study: DailyDialog vs. EmoWoZ



- **H1:** Strong alignment with existing emotion labels when using intermediate appraisal estimates.
- **H2:** Gradual and emotion-specific affect flow patterns which differ between corpora.
- Appraisals provide a latent structure across conversation corpora.

Key Contributions

- Introduced AppraisePLM: joint appraisal-emotion prediction model.
- Provides a scalable cross-corpus application of appraisal theory for conversation.
- Affective insight into emotion evolution in conversation.

Limitations and Future Directions

- Appraisal theory assumes retrospective evaluation; may differ in dialogue.
- Lack of appraisal-annotated conversation datasets.
- Need to optimize dimensionality
 - dimensions not linearly separable for non-qualitative trend analysis
 - different appraisal dimensions are more insightful for different emotions
- Framework for benchmarking empathetic agents
- Future: learn domain-specific appraisal dynamics, longitudinal affect tracking.

Thank You

Questions?

Code: github.com/alokdebnath/appraise-PLM

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